

SWAYAM Prompt Engineering - Week 2 Notes (Short with Examples)

Unlocking AI Precision

Generate → Evaluate → Refine. Improve prompts after checking accuracy and completeness.

Example: Prompt: 'Explain AI.' Refined: 'Explain AI for Class 12 students with 3 examples in 300 words.'

Summaries, Extraction & Translation

AI summarizes long text, extracts key facts and translates while preserving meaning.

Example: Summarize a 20-page paper, extract objectives and results, then translate into Telugu.

Summarisation & Structured Extraction

Convert unstructured text into organized tables, bullets or JSON.

Example: Extract a resume into Name, Skills, Education, Projects and Experience.

NotebookLM Audio Overview

Creates podcast-style discussions using only uploaded documents.

Example: Upload DBMS notes and generate a 10-minute revision podcast.

Mind Maps & Infographics

Turns documents into visual summaries like mind maps and infographics.

Example: Upload an AI chapter to create a mind map linking ML, DL and NLP.

NotebookLM Research Assistant

Summarizes papers, compares studies and finds research gaps.

Example: Compare three quantum computing papers by objective, dataset and results.

Evaluating & Refining AI Responses

Check AI output for accuracy, relevance and clarity, then improve prompts.

Example: Ask AI to fact-check and rewrite an answer with reliable sources.

Spreadsheet to Story (Julius.ai)

Analyzes Excel/CSV files, creates charts, dashboards and presentations.

Example: Upload monthly sales data to identify trends and generate a dashboard.